

Indian Institute of Technology Jodhpur

Class Test-2

Mathematics-I (MAL1010)

Date: November 03, 2025

Marks:20

Duration: 1 Hour

1. Prove that if a function $f : [a, b] \rightarrow \mathbb{R}$ is **continuous** on a closed and bounded interval $[a, b]$, then f is integrable on $[a, b]$.

(3)

2. Prove or disprove the following statements: (If you choose to disprove any of the following statements, you may do so by providing a counterexample.)

(a) If $f : [a, b] \rightarrow \mathbb{R}$ is **integrable** on $[a, b]$ and $f(x) > 0$ for $x \in [a, b]$, then $\frac{1}{f}$ is integrable on $[a, b]$.

(b) If $f : [a, b] \rightarrow \mathbb{R}$ is **continuous** on $[a, b]$ and $f(x) > 0$ for all $x \in [a, b]$, then $\frac{1}{f}$ is not integrable on $[a, b]$.

(c) If $f : I \rightarrow \mathbb{R}$ is integrable on a closed and bounded interval I and $g : J \rightarrow \mathbb{R}$ is integrable on a closed and bounded interval J , then $g \circ f$ is integrable on I .

(2 × 3 = 6)

3. Answer the following:

(a) Give an example of a function having infinitely many discontinuities and not being integrable.

(b) Give an example of a function having infinitely many discontinuities but being integrable.

(2)

4. Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

(2)

5. Find the volume of the solid generated by revolving the region between the parabola $x = y^2 + 1$ and the line $x = 3$ about the line $x = 3$.

(3)

6. Let

$$f(x, y) = x + y + \frac{xy}{\sqrt{x^2 + y^2}}, \quad x > 0, y > 0.$$

Discuss the existence of double limits and repeated limits of the function at $(0, 0)$.

(2)

7. Examine the continuity at the origin for:

$$f(x, y) = \begin{cases} xy & \text{if } |x| \geq |y| \\ -xy & \text{if } |x| < |y| \end{cases}$$

(2)

Indian Institute of Technology Jodhpur

Class Test-3

Mathematics-I (MAL1010)

Date: November 15, 2025

Marks:20

Duration: 1 Hour

1. Let

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq (0, 0), \\ 0, & (0, 0) \end{cases}$$

Show that (i) $f(x, y)$ is continuous at $(0, 0)$, (ii) $f(x, y)$ has partial derivatives at $(0, 0)$, and (iii) $f(x, y)$ is not differentiable at $(0, 0)$.

$$(1 + 2 + 3 = 6)$$

2. Find all the maxima and minima of the function

$$f(x, y) = x^3 + y^3 - 63(x + y) + 12xy.$$

$$(5)$$

3. Evaluate the following integrals:

(a) $\iint_R xy(x^2 + y^2) dx dy$ over $R = [0, a] \times [0, b]$.

(b) $\iiint_R x^3 z^4 (y^2 + 1) dx dy dz$ over $R = [0, 1] \times [0, 1] \times [0, 1]$.

(c) $\iint_R (x^2 + y^2) dx dy$, where R is bounded by $y = x^2$, $x = 2$, and $y = 1$.

$$(3 + 3 + 3 = 9)$$

Indian Institute of Technology Jodhpur

Minor

Mathematics-I (MAL1010)

Date: September 18, 2025

Marks: 35

Duration: 2 Hours

1. (a) Let S be a bounded subset of $\mathbb{R}$ with $\sup S = M$ and $\inf S = m$. Prove that the set $T = \{|x - y| : x \in S, y \in S\}$ is a bounded set and $\sup T = M - m$
(b) Check whether the following function is bounded or not on $(0, 1)$. Justify your answer.

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{if } x > 0, \\ 0 & \text{if } x = 0. \end{cases}$$

(3+2=5)

2. (a) Show that the series

$$\sum \frac{1}{n^p}$$

diverges for $p \leq 1$.

- (b) Prove that an absolutely convergent series is convergent.
(c) Test the convergence of the following series:

(i)

$$\frac{1 \cdot 2}{3^2 \cdot 4^2} + \frac{3 \cdot 4}{5^2 \cdot 6^2} + \frac{5 \cdot 6}{7^2 \cdot 8^2} + \dots$$

(ii)

$$\frac{1+2}{2^3} + \frac{1+2+3}{3^3} + \frac{1+2+3+4}{4^3} + \dots$$

(3+2+(2+2)=9)

3. (a) Evaluate the following limits with proper justifications:

(i) $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$

(ii) $\lim_{x \rightarrow 0} [x]$ where $[x]$ is the greatest integer value not greater than x .

- (b) Let $D \subset \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a function. Let c be a limit point of D . If $\lim_{x \rightarrow c} f(x) = l \in \mathbb{R}$, then f is bounded on some neighbourhood of c .

((2.5+2.5)+3=8)

4. Find the limit of the following sequences. Justify your answers:

(a) $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+n}} \right)$

(b) $\lim_{n \rightarrow \infty} (\sqrt{n^2+3n} - n)$.

(c) $\lim_{n \rightarrow \infty} \frac{\sin n}{n}$

(2+2+2=6)

5. (a) Let a function $f : [0, 1] \rightarrow \mathbb{R}$ is continuous on $[0, 1]$ and $f(x) \in \mathbb{Q}$, $\forall x \in [0, 1]$. If $f\left(\frac{1}{2}\right) = \frac{1}{2}$, prove that $f(x) = \frac{1}{2}$, for all $x \in [0, 1]$.

P.T.O.

(b) A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(0) = 0$ and

$$f(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ \frac{1}{q}, & \text{if } x = \frac{p}{q}, \text{ where } p \in \mathbb{Z}, q \in \mathbb{N} \text{ and } \gcd(p, q) = 1 \end{cases}$$

Show that f is not differentiable at 0.

(c) Use the MVT to prove that: $|\sin a - \sin b| \leq |a - b|$, for all $a, b \in \mathbb{R}$

(3+2+2=7)

Indian Institute of Technology Jodhpur

Major

Mathematics-I (MAL1010)

Date: **November 26, 2025**

Marks: **80**

Duration: **3 Hours**

1. (a) If $f(x) = |x|$ and $g(x) = 2|x|$, show that $f(x)$ and $g(x)$ are not differentiable at $x = 0$. But $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ exists and is equal to $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$.
- (b) Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Prove that if $f'(x) = 0$ for all $x \in [a, b]$, then f is a constant function on $[a, b]$.

((3+2)+3=8)

2. (a) Let a function $f : [a, b] \rightarrow \mathbb{R}$ be monotone on $[a, b]$. Then prove that f is integrable on $[a, b]$. Is the converse of this theorem true? Justify your answer.
- (b) The region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ is revolved about the x-axis to generate a solid. Find the volume of the generated solid.
- (c) Prove or disprove the following: If $f : [a, b] \rightarrow \mathbb{R}$ is integrable on $[a, b]$ and $f(x) > 0$ for $x \in [a, b]$, then $\frac{1}{f}$ is integrable on $[a, b]$.

((4+1)+3+2=10)

3. (a) Let

$$f(x, y) = \begin{cases} x \sin\left(\frac{1}{y}\right) + \frac{x^2 - y^2}{x^2 + y^2}, & \text{if } y \neq 0, \\ 0, & \text{if } y = 0. \end{cases}$$

Show that $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y)$ exists, but $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ and $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y)$ do not exist.

- (b) Show that the following function f is continuous at $(0, 0)$:

$$f(x, y) = \begin{cases} x \sin \frac{1}{y} + y \sin \frac{1}{x}, & xy \neq 0, \\ 0, & xy = 0 \end{cases}$$

- (c) Show that the following function f is not continuous at $(0, 0)$:

$$f(x, y) = \begin{cases} \frac{x^4 + y^4}{x - y}, & x \neq y, \\ 0, & x = y. \end{cases}$$

(5+3+2=10)
P.T.O....

4. (a) Let

$$f(x, y) = (|xy|)^{\frac{1}{2}}$$

Show that both f_x and f_y exist at $(0, 0)$ but the function f is not differentiable at $(0, 0)$

- (b) Let

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & \text{otherwise} \end{cases}$$

Show that $f_{xy} \neq f_{yx}$ at $(0, 0)$.

(5+5=10)

5. (a) Find stationary points of $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ and examine whether they are maxima or minima.

- (b) Solve the problem

$$\begin{aligned} \min \quad & x \log(x) + y \log(y) \\ \text{subject to} \quad & x + y = 10. \end{aligned}$$

Also verify second-order condition.

(5+5=10)

6. (a) Let $D := \{(x, y, z) \in \mathbb{R}^3 : x \geq 0, y \geq 0, z \geq 0, x \leq y + z \text{ and } 1 \leq 2(x + y + z) \leq 2\}$, and $f(x, y, z) = \frac{y+z}{x+y+z}$. Find the triple integral of f over D .

- (b) For $a > 0$ suppose $D := \{(x, y, z) \in \mathbb{R}^3 : a^2 \leq x^2 + y^2 + z^2 \leq 2az\}$. Consider $f : D \rightarrow \mathbb{R}$ defined by $f(x, y, z) := z$. Find the triple integral of f over D .

- (c) Using double integration, find the area of the triangle with vertices $(0, -1), (1, 0), (0, 1)$. (No marks for using other methods).

(4+4+2=10)

7. (a) Express $\vec{B} = 4\hat{i} - 8\hat{j} + 12\hat{k}$ as sum of a vector parallel to $\vec{A} = \hat{i} + 2\hat{j} - \hat{k}$ and a vector orthogonal to $\vec{A}$.

- (b) Suppose $\vec{F} = (y + z)\hat{i} + (z + x)\hat{j} + (x + y)\hat{k}$. Find work done by force $\vec{F}$ from $(0, 0, 0)$ to $(1, 1, 1)$ by line segment joining $(0, 0, 0)$ to $(1, 1, 0)$ followed by line segment joining $(1, 1, 0)$ to $(1, 1, 1)$.

- (c) Find the maximum value of the directional derivatives of $\phi = x^2 + z^2 - y^2$ at the point $(1, 2, 3)$. Find also the direction in which it occurs.

(3+3+(3+1)=10)

8. (a) Given $\phi = x^2yz$ and $\hat{A} = 2xz\hat{i} - yz\hat{j} + 3xz^3\hat{k}$, find $\text{curl}(\phi\hat{A})$ and $\text{curl}(\text{curl}\hat{A})$.

- (b) Using Green's theorem, find

$$\oint_C 3x^2y^2dx + 2x^2(1 + xy)dy$$

where C is the circle with center $(2, 0)$ and radius 2.

- (c) Find the outward flux of $\vec{F} = x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ over the region cut from the solid cylinder $x^2 + y^2 \leq 4$ by planes $z = 0$ and $z = 1$.

((2+2)+4+4=12)



Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II MAL1020
Quiz II

AY 25-26

Semester II

Full Marks 10

Duration: 45 Mins

1. Diagonalize the following matrix *orthogonally*. $A = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 5 & 2 \\ 1 & 2 & 2 \end{pmatrix}$. (2)

For finding eigenvalues and linearly independent eigenvectors, 1 mark and for a correct implementation of the Gram-Schmidt process, remaining 1 mark.

2. Solve the initial value problem

$$y' = \frac{2xy - \sin x - 3y^2 e^x}{6ye^x - x^2}, \quad y(0) = 2.$$

(2)

No part marking for this problem.

3. Solve the following ordinary differential equation

$$(x^2 + y^2 + 1)dx + (xy + y)dy = 0.$$

(2)

For finding the correct integrating factor, 1 mark and remaining 1 mark for the correct final answer.

4. Find the class of 2×2 matrices satisfying

$$A^2 - 5A + 6 = 0$$

(2)

There is no part marking for this question, and a full mark will be given only if you are using the concept of diagonalizability and similar matrices.

5. Let $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$.

- (a) Find A^{-1} using Cayley-Hamilton theorem.
(b) Express A^{-5} as a linear combination of I and A .

(1+1)

Best Wishes



Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II MAL1020
Quiz III

AY 25-26

Semester II

Maximum Marks 10

Duration: 1 Hr

1. Solve $xy' + y = y^2$.

(2)

No part marking for this problem.

2. Find first two Picard iteration for the initial value problem

$$y' = x^2 + y^2, y(-1) = 1.$$

(2)

For each correct iteration, 1 mark.

3. Find the general solution for the following second-order ODE:

$$y'' + 2y' - 3y = -2e^x \cos x.$$

(2)

For finding general solution of the homogeneous part, 1 mark and for particular solution, remaining 1 mark.

4. Solve by the method of variation of parameters:

$$u'' + 2u' + u = \frac{e^{-x}}{x^2}$$

(2)

No part marking for this problem.

5. Find two power series solutions about $x = 0$:

$$y'' + e^x y' - y = 0$$

(2)

1 mark for each correct power series.

6. Solve by power series method:

$$(x^2 + 1)y'' + 2xy' = 0, \quad y(0) = 0, y'(0) = 1$$

(2)

No part marking for this problem.

Best Wishes